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# Solving linear variational inequality problems by a self-adaptive projection method ☆

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**Abstract**

We propose a self-adaptive projection method for solving linear variational inequality problems and show its global convergence under mild conditions. Some numerical results are also addressed which indicate that the method is quite robust and efficient.


*Keywords:* Linear variational inequality problems; Projection and contraction methods; Global convergence

## 1. Introduction

We consider the linear variational inequality problem, denoted by LVI$(\Omega, M, q)$ for simplicity, which is to find a vector $x^* \in \Omega$ such that

$$(x - x^*)^\top (Mx^* + q) \geqslant 0, \quad \forall x \in \Omega, \tag{1}$$

where $\Omega$ is a nonempty, closed, convex subset of $R^n$, $M \in R^{n\times n}$ is a matrix and $q \in R^n$ is a given vector. When $\Omega = R^n_+$ (the nonnegative orthant of $R^n$), LVI$(\Omega, M, q)$ reduces to the linear complementarity problem LCP$(M, q)$ of finding $x^* \in R^n$, such that

$$x^* \geqslant 0, \quad Mx^* + q \geqslant 0, \quad (Mx^* + q)^\top x^* = 0. \tag{2}$$

The linear variational inequality problem and linear complementarity problem have a number of important applications in operations research, engineering problems and economic equilibrium problems.

It is well known [3] that solving LVI$(\Omega, M, q)$ is equivalent to solving the system of projection equations

$$x = P_\Omega[x - \beta(Mx + q)],$$

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where $\beta > 0$ is a positive constant and $P_\Omega[u]$ is the orthogonal projection of a vector $u$ from $R^n$ onto $\Omega$. Let

$$e(x,\beta) := x - P_\Omega[x - \beta(Mx+q)],$$

then solving LVI (1) is also equivalent to finding a zero point of $e(x,\beta)$ for a given constant $\beta > 0$.

There are many methods for LVI$(\Omega, M, q)$, see for example [2,5,7–18] and the references therein. Among these methods, the projection and contraction methods are attractive for their simplicity and efficiency. In [9,10], based on the inequality

$$(x-x^*)^\top (I+\beta M^\top)e(x,\beta) \geqslant \|e(x,\beta)\|^2, \tag{3}$$

He proposed a class of projection methods and proved that the distance between the iterative point to the solution set, denoted by $S$, is contracted. That is, the iterative sequence $\{x^k\}$ generated by the recursion

$$x^{k+1} = x^k - \gamma\alpha_k G^{-1}(I+\beta M^\top)e(x^k,\beta)$$

satisfies

$$\mathrm{dist}^2_G(x^{k+1}, S) \leqslant \mathrm{dist}^2_G(x^k, S) - c_0\|e(x^k,\beta)\|^2,$$

where

$$\mathrm{dist}_G(x,S) = \min\{\|x-x^*\|_G | x^* \in S\},$$

$\gamma \in (0,2)$, $c_0 > 0$ are two constants, $G \in R^{n\times n}$ is a symmetric positive definite matrix and

$$\alpha_k = \frac{\|e(x^k,\beta)\|^2}{e(x^k,\beta)^\top (I+\beta M)G^{-1}(I+\beta M^\top)e(x^k,\beta)}.$$

When $M$ is symmetric, He suggested to choose $G = I + \beta M$ and when $M$ is asymmetric, $G = (I+\beta M^\top)(I+\beta M)$.

Many applications have shown that if the parameter $\beta$ is set to be either too small or too large, the efficiency of the method will be depredated greatly. To choose a (sequence of) suitable parameter(s) is thus important. Recently, for the symmetric linear variational inequality problem, Liao and Wang proposed a self-adaptive projection method [13]. The numerical results reported there indicate that the method is robust and efficient.

However, in practical applications, the underlying matrix $M$ may not be symmetric. In this paper, by choosing $G = (I+\beta M^\top)(I+\beta M)$, we give a new self-adaptive projection method for linear variational inequality problems. The advantages of this choice is two folds:

1. It can also solve asymmetric linear variational inequality problems.
2. It can introduce computational efficiency.

Under the same assumptions as those in [13], we show the global convergence of the proposed method. Some computational experience reported in Section 3 show that the method has improved performance over the existed projection methods with fixed parameter [9,10] or variable parameters [13].

## 2. The algorithm and its convergence

We now state our method formally and then analyze its convergence.

**Algorithm 2.1.** A self-adaptive projection method.

S0. Choose $x^0 \in R^n$, $\gamma \in (0,2)$, $\beta_0 > 0$, $\epsilon \geqslant 0$, and a sequence $\{\tau_k\} \subseteq [0,\infty)$ with $\sum_{k=0}^{\infty}\tau_k < \infty$. Set $k := 0$.

S1. If

$$\|e(x^k,\beta_k)\|_\infty \leqslant \epsilon,$$

then stop; else go to the next step.

S2. Compute the next iterate by

$$x^{k+1} = x^k - \gamma(I+\beta_k M)^{-1}e(x^k,\beta_k). \tag{4}$$

S3. Choose the next parameter $\beta_{k+1}$

$$\frac{1}{1+\tau_k}\beta_k \leqslant \beta_{k+1} \leqslant (1+\tau_k)\beta_k. \tag{5}$$

Set $k := k+1$ and go to Step 1.

**Remark 2.1.** This method is just a modification of the method of He [9,10] where we get a 'suitable' parameter automatically. It can also be viewed as an extension of the method of Liao and Wang [13] by choosing a different symmetric matrix $G$ to solve nonsymmetric LVI.

**Remark 2.2.** In Step 2, we have to get the inverse of the positive definite matrix $I + \beta_k M$, which seems to be time consuming. However, in our numerical experience, it is enough to update $\beta_k$ a little times (not every iterate step), and it thus seems not to be a computational burden.

We now show the global convergence of the method, beginning with the following lemmas.

**Lemma 2.1.** *Suppose that $\tau_k \geqslant 0$ and $\sum_{k=0}^{\infty}\tau_k < \infty$, then there exist two constants $0 < \beta_{\min} \leqslant \beta_{\max} < \infty$, such that for all $k \geqslant 0$*

$$\beta_{\min} \leqslant \beta_k \leqslant \beta_{\max}.$$

**Proof.** Let

$$C_p := \prod_{k=0}^{\infty}(1+\tau_k), \quad C_s := \sum_{k=0}^{\infty}\tau_k.$$

Then it follows from the fact that $\tau_k \geqslant 0$, $C_s = \sum_{k=0}^{\infty}\tau_k < \infty$ that $C_p < \infty$. From (5) we have that

$$\{\beta_k\} \subset [\beta_0/C_p, C_p\beta_0].$$

The assertion then follows immediately with $\beta_{\min} = \beta_0/C_p$ and $\beta_{\max} = \beta_0 C_p$. □

**Lemma 2.2.** *For any $x \in R^n$ and $0 < \beta_1 \leqslant \beta_2$, we have*

$$\|e(x,\beta_1)\| \leqslant \|e(x,\beta_2)\|. \tag{6}$$

**Proof.** See Lemma 1 of [4] and (2.6) of [15]. □

It follows from Lemmas 2.1 and 2.2 that

$$\|e(x^k,\beta_{\min})\| \leqslant \|e(x^k,\beta_k)\| \leqslant \|e(x^k,\beta_{\max})\|.$$

Thus, if the algorithm stops at Step 1, then $x^k$ is an approximate solution of LVI$(\Omega, M, q)$. In the following, we assume that the algorithm generates an infinite sequence of iterate $\{x^k\}$.

We also use the following lemma due to Cheng [1].

**Lemma 2.3.** *Let $\{a^k\}$ and $\{b^k\}$ be two sequences of nonnegative numbers, with $\sum_{k=0}^{\infty}b^k < \infty$ and $a^{k+1} \leqslant a^k + b^k$. Then $\{a^k\}$ converges.*

**Theorem 2.4.** *Suppose that $M$ is positive semidefinite and the solution set $S$ of LVI$(\Omega, M, q)$ is nonempty. Let $\{x^k\}$ be the sequence generated by Algorithm 2.1 and $G_k := (I + \beta_k M^\top)(I + \beta_k M)$. Then the generated sequence $\{x^k\}$ converges to a solution to LVI$(\Omega, M, q)$ globally.*

**Proof.** Let $x^* \in S$ be an arbitrary solution of LVI$(\Omega, M, q)$. From (4) we have

$$\begin{aligned}
\|x^{k+1} - x^*\|_{G_k}^2 &= \|x^k - x^* - \gamma(I+\beta_k M)^{-1}e(x^k,\beta_k)\|_{G_k}^2 \\
&= \|x^k - x^*\|_{G_k}^2 - 2\gamma(x^k - x^*)^\top G_k (I+\beta_k M)^{-1}e(x^k,\beta_k) + \gamma^2\|(I+\beta_k M)^{-1}e(x^k,\beta_k)\|_{G_k}^2 \\
&= \|x^k - x^*\|_{G_k}^2 - 2\gamma(x^k - x^*)^\top (I+\beta_k M^\top)e(x^k,\beta_k) + \gamma^2\|e(x^k,\beta_k)\|^2 \\
&\leqslant \|x^k - x^*\|_{G_k}^2 - \gamma(2-\gamma)\|e(x^k,\beta_k)\|^2, 
\end{aligned} \tag{7}$$

where the inequality follows from (3). On the other hand, it follows from (5) that

$$\|x^{k+1}-x^*\|^2_{G_{k+1}} = \|(I+\beta_{k+1}M)(x^{k+1}-x^*)\|^2 \leqslant \|(1+\tau_k)(I+\beta_k M)(x^{k+1}-x^*)\|^2 = (1+\tau_k)^2\|x^{k+1}-x^*\|^2_{G_k}, \tag{8}$$

which, together with (7), means that

$$\|x^{k+1}-x^*\|^2_{G_{k+1}} \leqslant (1+\tau_k)^2\|x^k-x^*\|^2_{G_k} - \gamma(2-\gamma)\|e(x^k,\beta_k)\|^2. \tag{9}$$

Thus,

$$\|x^{k+1}-x^*\|^2_{G_{k+1}} \leqslant (1+\tau_k)^2\|x^k-x^*\|^2_{G_k} \leqslant \cdots \leqslant \left(\prod_{i=0}^{k}(1+\tau_i)^2\right)\|x^0-x^*\|^2_{G_0}.$$

Since $\{\beta_k\}$ is bounded, it follows from the above inequality that $\{x^k\}$ is also bounded. Also from (9) we have that

$$\gamma(2-\gamma)\sum_{k=0}^{\infty}\|e(x^k,\beta_k)\|^2 \leqslant \sum_{k=0}^{\infty}(2\tau_k+\tau_k^2)\|x^k-x^*\|^2_{G_k} + \|x^0-x^*\|^2_{G_0} < \infty.$$

Using Lemma 2.2 and the above inequality we have

$$\lim_{k\to\infty}\|e(x^k,\beta_{\min})\| \leqslant \lim_{k\to\infty}\|e(x^k,\beta_k)\| = 0.$$

Since $\{x^k\}$ is bounded, it has at least one cluster point. Let $\tilde{x}$ be a cluster point of $\{x^k\}$ and $x^{k_j}$ be the subsequence converging to $\tilde{x}$. Then, taking limit along the subsequence and using the continuity of $e(\cdot,\beta)$, we have that

$$e(\tilde{x},\beta_{\min}) = \lim_{j\to\infty} e(x^{k_j},\beta_{\min}) = 0,$$

which means $\tilde{x}$ is a solution of LVI$(\Omega,M,q)$. Since $x^*\in S$ is an arbitrary solution, we can just set $x^* := \tilde{x}$ in the above analysis and

$$\|x^{k+1}-\tilde{x}\|^2_{G_{k+1}} \leqslant \|x^k-\tilde{x}\|^2_{G_k} + (2\tau_k+\tau_k^2)\|x^k-\tilde{x}\|^2.$$

Using the fact that $\{x^k\}$ is bounded, $\tau_k \geqslant 0$ and $\sum_{k=0}^{\infty}\tau_k < \infty$ and Lemma 2.3, we can conclude that the whole sequence $\{x^k\}$ converges to $\tilde{x}$, a solution of LVI$(\Omega,M,q)$. This completes the proof. □

## 3. Computational results

In this section, we present some numerical results for the proposed self-adaptive projection method. Our main interests are two folds: the fist one is in comparing the proposed method with Liao and Wang's method [13], showing the numerical advantage; the second one is in showing that it can also solve nonsymmetric linear variational inequality problems.

For the sake of balance, it is reasonable to hope that

$$\|x^{k+1}-x^k\| \approx \|\beta_k M(x^{k+1}-x^k)\|$$

or

$$\|e(x^k,\beta_k)\| \approx \|\beta_k Me(x^k,\beta_k)\|.$$

This consideration provides a way of choosing the parameter $\beta$ and Liao and Wang [13] propose the following strategy. Let

$$\omega_k = \frac{\|\beta_k Me(x^k,\beta_k)\|}{\|e(x^k,\beta_k)\|}, \tag{10}$$

then, let

$$\beta_{k+1} = \begin{cases} (1+\tau_k)\beta_k & \text{if } \omega_k < \frac{1}{(1+\mu)}, \\ \frac{1}{(1+\tau_k)}\beta_k & \text{if } \omega_k > 1+\mu, \\ \beta_k & \text{otherwise,} \end{cases} \tag{11}$$

where $\mu > 0$ is a given constant.

The test problem considered here is the linear complementarity problem (2), a special case of (1) with $\Omega = R^n_+$. The projection onto $\Omega$ in the sense of Euclidean-norm is very easy to carry out. For any $y \in R^n$, $P_\Omega[y]$ is defined componentwise as

$$(P_\Omega[y])_j = \begin{cases} y_j & \text{if } y_j \geqslant 0, \\ 0 & \text{otherwise.} \end{cases}$$

In our first test problem we form the matrix $M$ similarly as in [6]. The matrix $M = A^\top A$, where $A$ is an $n \times n$ matrix whose entries are randomly generated in the interval $(-5, +5)$. The vector $q$ is generated from a uniform distribution in the interval $(-500, +500)$ or $(-500, 0)$.

We use the proposed self-adaptive projection method to solve this set of problems and test the proposed method with the parameter adjustment strategy (10) and (11), $\tau_k$ is set to 0.5 if the adjustment times is less than 30 and 0 otherwise, and $\mu = 1.0$ for all problems. For comparison purpose, we also coded the self-adaptive method of Liao and Wang [13] using the same adjustment strategy.

All codes were written in Matlab and run on a P-III 600 personal computer. In all test examples we take $\gamma = 1.9$. The iterations begin with $x^0 \in R^n$ generated randomly in $(0, 1)$ and stop as soon as $\|e(x^k, 1)\|_\infty \leqslant 10^{-7}$. The number of iterations and the computation times of Liao and Wang's method (LW method) and the proposed method for the problem with different sizes are given in Tables 1 and 2. In these tables, 'IN' means the number of iterations, 'AN' means the number of adjustments and 'CPU' means the cputime in seconds.

The results show that the two self-adaptive methods both converge to the solution for each test problem. For most initial parameter $\beta_0$, the number of iterations and the CPU time of the proposed algorithm are less than the method of Liao and Wang [13].

To show that the proposed method can also solve asymmetric linear complementarity problem, we test it with $M = A^\top A + B$ where the entries of the $n \times n$ matrix $A$ are randomly generated in the interval $(-5, +5)$ and a skew-symmetric matrix $B$ is generated in the same way. Table 3 reports the numerical results with $n$ varying from 100 to 1000 and $\beta_0 = 0.01$.

The results in Table 3 indicate that the proposed method is quite robust and efficient, and the number of iterations is relatively independent of the problem sizes.

Table 1
Numerical results for $n = 100$ and $q \in (-500, 500)$

| $\beta$ | Proposed method | | | LW method | | |
|---|---|---|---|---|---|---|
| | IN | AN | CPU (s) | IN | AN | CPU (s) |
| $10^{-6}$ | 76 | 13 | 0.33 | 322 | 13 | 0.83 |
| $10^{-5}$ | 116 | 7 | 0.38 | 318 | 10 | 0.88 |
| $10^{-4}$ | 91 | 4 | 0.27 | 317 | 5 | 0.88 |
| $10^{-3}$ | 68 | 3 | 0.16 | 319 | 2 | 0.83 |
| $10^{-2}$ | 114 | 3 | 0.33 | 315 | 4 | 0.88 |
| $10^{-1}$ | 83 | 6 | 0.27 | 310 | 7 | 0.88 |
| 1 | 81 | 11 | 0.33 | 314 | 14 | 0.82 |
| 10 | 131 | 13 | 0.50 | 318 | 11 | 0.88 |
| 100 | 101 | 16 | 0.44 | 323 | 17 | 0.87 |
| 1000 | 126 | 20 | 0.55 | 326 | 23 | 0.93 |

Table 2
Numerical results for $n = 300$ and $q \in (-500, 0)$

| $\beta$ | Proposed method | | | LW method | | |
|---|---|---|---|---|---|---|
| | IN | AN | CPU (s) | IN | AN | CPU (s) |
| $10^{-6}$ | 96 | 9 | 10.39 | 362 | 10 | 13.68 |
| $10^{-5}$ | 78 | 6 | 7.64 | 360 | 7 | 13.57 |
| $10^{-4}$ | 119 | 4 | 6.26 | 355 | 3 | 13.41 |
| $10^{-3}$ | 95 | 3 | 6.15 | 359 | 3 | 13.57 |
| $10^{-2}$ | 78 | 6 | 7.58 | 355 | 7 | 13.40 |
| $10^{-1}$ | 123 | 8 | 10.60 | 347 | 9 | 13.08 |
| 1 | 107 | 13 | 12.30 | 350 | 12 | 15.05 |
| 10 | 93 | 16 | 13.22 | 358 | 15 | 14.23 |
| 100 | 119 | 18 | 14.05 | 355 | 19 | 13.46 |
| 1000 | 126 | 21 | 16.10 | 362 | 22 | 13.85 |

Table 3
Numerical results for asymmetric LCP with $q \in (-500, 0)$

| $n$ | 100 | 150 | 200 | 250 | 300 | 350 | 400 | 500 | 800 | 1000 |
|---|---|---|---|---|---|---|---|---|---|---|
| IN | 89 | 112 | 115 | 144 | 150 | 161 | 125 | 153 | 149 | 115 |
| AN | 9 | 8 | 10 | 9 | 9 | 8 | 7 | 8 | 9 | 9 |
| CPU | 0.33 | 1.10 | 2.80 | 6.86 | 12.35 | 16.15 | 23.01 | 62.67 | 152.17 | 253.78 |

## 4. Concluding remarks

This paper gives a self-adaptive projection method for solving symmetric and asymmetric linear variational inequality problems. Under suitable conditions, the method was shown to converge to a solution of the problem globally. The limited computational results show that the proposed method is quite stable and efficient and have some advantages over the same type of methods [9,10,13].

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